

CSCI 264 — Assembly Language

Lab 3

Pointers and Dynamic Memory in C

Write a C program, `stats.c`, which reads a count `n`, then `n` integers and stores them in an array. Then find their minimum and maximum values and swap the positions in the array.

Store the values in an array allocated with `malloc`, since the size is not known until the program runs. Check the value `malloc` returns for `NULL`, and `free` the array before the program ends. You may assume `n` is at least 1.

Functions

You must write each of these functions:

```
void min_max(int *arr, int size, int *min, int *max);
void swap(int *a, int *b);
```

- `min_max` stores the location of the smallest element in `*min` and the location of the largest in `*max`. It does not return anything; `main` gets both answers through the pointer parameters.
- `swap` exchanges the two values at positions `a` and `b`.

Memory Addresses

Print the following memory addresses inside the functions using `%p`:

`main()`

- the array: both `&arr` and `arr`

`min_max()`

- the `arr` parameter: both `&arr` and `arr`

`swap()`

- `&a` and `a`

Label each line with the function name and what is printed, like this:

```
printf("[main]    &n    = %p\n", &n);
printf("[min_max] arr    = %p\n", arr);
```

Test case:

n	Values	Min	Max	Swapped
5	4 -2 0 9 7	-2	9	4 9 0 -2 7

Submission

Submit your source file on Canvas by Tuesday, September 22 and demo the program to Professor West by the end of the next lab session. Put your name in a comment at the top of the source file. If a classmate helped you put their name in the comment. Disclose any use of Generative AI in a comment at the top.